

Authorized to Decrypt, Not to Infer: Stateful Disclosure for Research Knowledge Graphs

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Abstract—An authorized recipient can infer a protected fact without ever decrypting unauthorized data. In a research knowledge graph, two replies that are harmless independently may expose a hidden relationship when combined with prior disclosures, ontology rules, or external knowledge. Attribute-based encryption and per-release redaction cannot prevent this mosaic attack because every released fragment may be individually authorized. We model cross-laboratory sharing as an adaptive interaction between a disclosure agent and an information-seeking requester, treating the accumulated observable transcript rather than an individual subgraph as the security object. We develop a stateful disclosure gate that generates task-relevant candidate views, rejects views that violate logical opacity, and meters posterior-odds amplification across repeated requests and declared coalitions. Approved views pass through an existing encrypted selective-disclosure proof of concept. The evaluation separates exhaustive micro-world evidence, generated laboratory episodes with oracle and reconstructed graphs, and end-to-end delivery checks. It measures cumulative leakage, attack success, task utility, and robustness to incomplete background models. *[Insert one numerical headline and one bounded conclusion after the experiments are frozen and run.]*

Index Terms—knowledge graphs, inference control, controlled query evaluation, selective disclosure, adaptive privacy, personal agents

I. INTRODUCTION

Lab A authorizes Lab B to receive two answers. The first says that Alice reserved instrument I . The second says that Project Orchid is the only project using I this week. Neither answer states that Alice works on Orchid. Taken together with a public rule connecting reservations to the sole project using an instrument, however, they disclose exactly that protected relationship. No ciphertext was broken. Lab B was entitled to decrypt both answers. The leak occurred *after* access control had done its job.

Figure 1 is the smallest instance of a broader problem. Knowledge graphs are valuable because facts compose: shared entities, paths, ontologies, and correlations turn fragments into new knowledge. The same multiplicative property makes independent release decisions unsafe. A requester can remember prior answers, ask overlapping questions, interpret refusals, join public information, and collaborate with another identity. Consequently, the security object is the complete observable history, not one query, one graph fragment, or one ciphertext.

This paper belongs to a three-question research program. Q1 asks how personal agents transform member mailboxes and records into provenance-bearing personal graphs and how a lab agent forms a laboratory graph. Q2 asks how an approved fragment crosses a laboratory boundary; an

existing proof of concept provides multi-authority attribute-gated encryption, content-addressed storage, and append-only provenance metadata. Q3 asks what Q1 and Q2 do not answer: how can the publisher know that the *combination* of authorized disclosures does not reveal protected knowledge?

We answer Q3 by modeling the exchange as a stateful two-agent game. Agent A owns a laboratory graph and wants to complete a legitimate task with the least collateral inference. Agent B is authorized but information-seeking: it chooses later requests from earlier observations and attempts to infer more than any single answer states. A stateful gate sits between graph construction and encrypted delivery. It proposes exact, minimized, generalized, and refusal responses; checks each complete observable response against the accumulated coalition history; and releases the highest-utility candidate that satisfies both a logical and a probabilistic confidentiality condition.

The paper makes three contributions:

- We formalize adaptive, authorized, longitudinal inference from knowledge-graph disclosures, including graph structure, prior answers, refusals, metadata, and declared collusion.
- We combine a hard logical-opacity condition with a transcript-level bound on posterior-odds amplification, and give a history-aware preservation result under explicit model assumptions.
- We define a figure-led evaluation that separates exact bounded evidence, generated-world privacy-utility evidence, and proof-of-concept delivery evidence. The design uses oracle ground truth to isolate disclosure security and reconstructed graphs to measure sensitivity to Q1 error.

II. FROM AUTHORIZED SHARING TO SEMANTIC LEAKAGE

A. Q1 as a Declared Graph Interface

For member i , a personal agent constructs a graph from the evidence visible through time t :

$$\hat{G}_i(t) = \text{Construct}(M_i^{\leq t}). \quad (1)$$

A lab agent aggregates these graphs while retaining source, ownership, time, confidence, and policy annotations:

$$\hat{G}_L(t) = \biguplus_i \text{Annotate}(\hat{G}_i(t), i, \pi_i, t). \quad (2)$$

This is an interface, not a claim that arbitrary email can be converted perfectly into truth. The companion Generation environment provides two experimental inputs: an oracle graph G_L^* exported from its authoritative world and epistemic

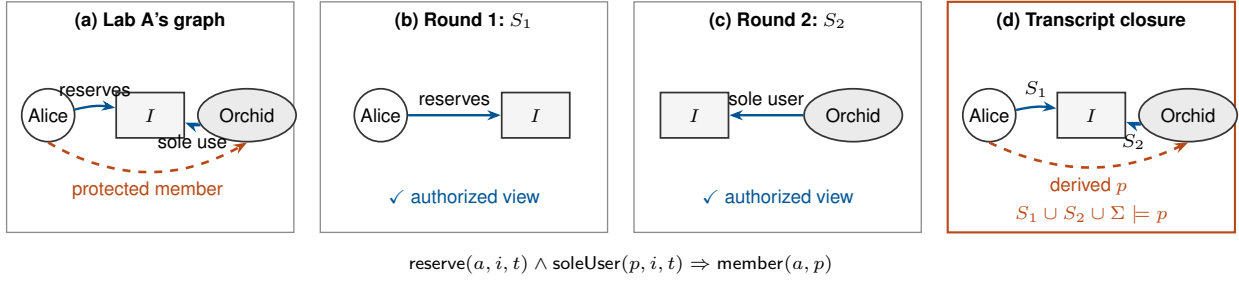


Fig. 1. Individually authorized disclosures can compose into a protected fact. Each fragment is harmless in isolation; their accumulated semantic closure is not. Encryption controls access to both fragments but cannot prevent the authorized inference.

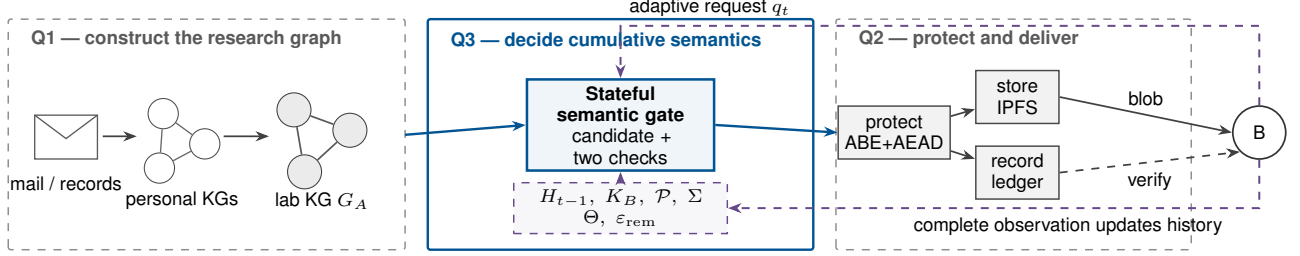


Fig. 2. System context and manuscript boundary. Q1 supplies an oracle or reconstructed laboratory graph. This paper contributes the highlighted Q3 gate, which decides what cumulative semantics may be released. The existing Q2 proof of concept stores the encrypted view and records separate verification metadata for the recipient.

state, and a reconstructed graph $\hat{G}_L$ built from the messages available to personal agents. Primary Q3 experiments use G_L^* so extraction error cannot masquerade as a disclosure failure. Secondary experiments replace it with $\hat{G}_L$ and report the difference.

B. Q2 as the Delivery Substrate

The existing Sharing proof of concept selects an RDF/JSON-LD fragment, encrypts it with AES-GCM, protects the symmetric seed under a multi-authority ciphertext-policy attribute formula, stores the encrypted object through IPFS, and records content and policy hashes with versioned provenance metadata. The cryptographic construction follows Rouselakis–Waters multi-authority ABE [1] through Charm [2]; IPFS supplies content-addressed storage [3].

That implementation supports a narrow claim: under its key and publisher assumptions, an authorized attribute set can recover the selected plaintext while an insufficient set cannot, and ciphertext or blob modification is detectable. It does not certify that the selected triples are semantically minimal, that a sequence of permitted views is inference-safe, or that identity, revocation, availability, and key custody are production-ready. We freeze Q2 as a delivery substrate and add only the evidence-producing components needed to study Q3.

C. The Authorized Adaptive Requester

Agent A controls a private graph G_A , a public or declared inference theory Σ , a protected policy $\mathcal{P}$, and a model of B's

background knowledge K_B . At round t ,

$$\begin{aligned} q_t &\leftarrow \pi_B(H_{t-1}, K_B), \\ y_t &\leftarrow \mathcal{M}_A(G_A, q_t, H_{t-1}), \\ H_t &= H_{t-1} \parallel (q_t, y_t). \end{aligned} \quad (3)$$

Here y_t denotes the *complete modeled observation*: released triples or text, generalization level, refusal or omission, response form and size, provenance and policy metadata, modeled timing, and any completeness claim. Treating only plaintext triples as observable would move the leak rather than close it.

B is authorized for every plaintext it receives. It knows the schema, public ontology, policy language, and release algorithm; remembers prior exchanges; and may use symbolic deduction, graph completion, Bayesian inference, differencing, and language-model reasoning. Declared experiments also allow transcript pooling across cooperating identities. Cryptographic breaks, compromised endpoints, poisoning of G_A , and unrestricted physical-world side channels are outside this paper's guarantee.

III. STATEFUL INFERENCE-AWARE DISCLOSURE

Figure 3 shows the per-round decision. An agent may interpret the task and propose responses, but it cannot authorize its own output. Explicit monitors evaluate the response that B will actually observe, using persistent state that cannot be reset by starting a new conversation.

A. Candidate Ladder and Task Utility

For request q_t , A builds a finite candidate set

$$\mathcal{C}_t = \{y_t^{\text{exact}}, y_t^{\text{minimal}}, y_t^{\text{generalized}}, y_t^{\text{aggregate}}, y_t^{\text{fallback}}\}. \quad (4)$$

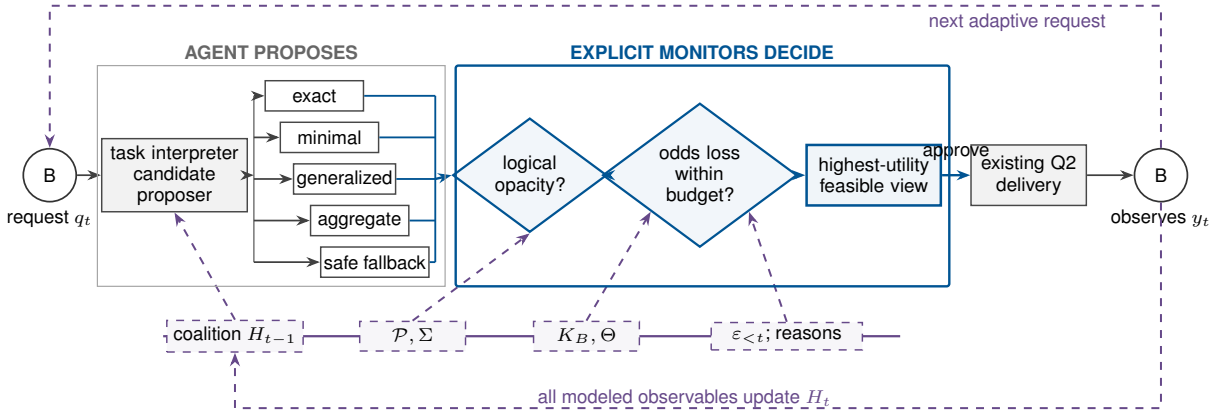


Fig. 3. Per-round disclosure decision and persistent security state. Candidate generation can be heuristic or agentic; explicit history-aware monitors make the release decision. Every modeled observation returns to the coalition transcript before the next request.

Candidate transformations may remove unnecessary triples, coarsen time or location, replace a named entity with a class, return a range or aggregate, delay a response for batching, or issue an observation-safe fallback. All compared mechanisms receive the same candidate pool and task-specific utility evaluator. This prevents an especially capable answer generator from being confused with an especially safe security mechanism.

Task utility is scored against an answer key for the legitimate request, not against fragment size alone. Candidate cost includes semantic specificity and excess disclosure after canonicalization, so a paraphrase does not appear safer merely because it uses different tokens.

B. History-Aware Security Gates

The logical gate evaluates the closure of the modeled background, prior transcript, and candidate response under a bounded rule language. It blocks candidates that newly entail a protected proposition or eliminate a declared protected alternative. Minimal derivations are retained so an unsafe candidate can be transformed rather than rejected blindly.

The probabilistic gate evaluates how strongly the same candidate changes the relative likelihood of protected alternatives under a declared family of background models. Each approved conditional loss is charged to a persistent ledger. The ledger is keyed to a declared coalition when collusion is in scope; a per-login history is not a meaningful defense against transcript pooling.

The highest-utility feasible response is

$$y_t^* = \arg \max_{y \in C_t} [U(q_t, y) - \lambda \text{Cost}(y)] \quad (5)$$

subject to both gates and the remaining cumulative budget. If no candidate is feasible, the mechanism uses a fallback whose observable form has itself passed the same checks. Only then does Q2 serialize, encrypt, store, and record the view.

IV. SECURITY FORMULATION

A. Declared Guarantee Boundary

Let $\mathcal{P}$ contain protected propositions or discriminative pairs (s_i, s_j) , such as presence versus absence of a relationship or two possible values of a derived attribute. Let Ω be the graphs consistent with A's declared background model and the public behavior of the mechanism. No useful mechanism can promise zero inference against arbitrary auxiliary information [4]; therefore every claim is relative to $\mathcal{P}$, Σ , K_B , Ω , the modeled observables, and an interaction horizon.

B. Logical Opacity

Definition 1 (Relative logical opacity). A transcript H_t is logically opaque for protected set $\mathcal{P}$ when

$$\text{Cn}_\Sigma(K_B \cup H_t) \cap \mathcal{P} = \text{Cn}_\Sigma(K_B) \cap \mathcal{P}. \quad (6)$$

The equality says that the exchange introduces no new protected entailment under the declared theory. When learning either truth or falsity must be prevented, both alternatives are represented in the policy. A stronger possible-world reading requires the observed behavior to remain compatible with at least one world under each protected alternative. That reading is important for refusals: a deterministic refusal may eliminate every world in one alternative even when its text contains no secret.

C. Transcript-Level Probabilistic Confidentiality

Logical ambiguity is insufficient when one surviving alternative becomes overwhelmingly likely. For every declared model $\theta \in \Theta$, secret pair (s_i, s_j) , and transcript event h , we require

$$e^{-\varepsilon} \leq \frac{\Pr_\theta[H_T = h \mid s_i]}{\Pr_\theta[H_T = h \mid s_j]} \leq e^\varepsilon. \quad (7)$$

For nonzero slack, the implementation uses the corresponding two-sided (ε, δ) inequalities rather than the ratio in Equation (7). When $\delta = 0$, Bayes' rule gives the operational meaning: observing the complete transcript changes prior odds between the alternatives by at most e^ε . The secret-pair and prior-family

formulation follows the intuition of Pufferfish privacy [5]; unlike a one-shot statement, our release rule is explicitly conditional on every feasible prior history because general Pufferfish mechanisms do not compose automatically [6].

D. History-Aware Preservation

Theorem 1 (Adaptive transcript preservation). *Assume H_0 satisfies Equation (6). At every round t , the controller emits the complete observation y_t only if the extended transcript preserves logical opacity and, uniformly over every feasible prior history and protected pair, its conditional likelihood ratio is at most e^{ε_t} . Then H_T remains logically opaque and satisfies Equation (7) with*

$$\varepsilon_{1:T} \leq \sum_{t=1}^T \varepsilon_t. \quad (8)$$

Proof sketch: logical opacity follows by induction because the actual extended transcript is checked before each release. For the probabilistic condition, factor the adaptive transcript likelihood into round-conditional terms. Applying the uniform conditional bound to each factor and multiplying yields $\exp(\sum_t \varepsilon_t)$. The condition must include every modeled output mode; otherwise a refusal or metadata choice becomes an unaccounted factor. Detailed proof obligations and the restricted rule language belong in the supplementary material.

The logical layer is grounded in dynamic Controlled Query Evaluation, where answers depend on the query history and confidentiality policy [7]–[9]. The probabilistic layer addresses correlated evidence that need not yield an exact proof. Neither layer substitutes for the other, as Figure 4 illustrates.

V. EXPERIMENTAL METHODOLOGY

The evaluation is organized around three questions:

- **RQ1—Attack:** How often can an authorized adaptive requester infer protected knowledge that was never explicitly released?
- **RQ2—Trade-off:** How much cumulative leakage does stateful control prevent, and what task utility and specificity does it cost?
- **RQ3—Boundary:** How do long histories, collusion, missing background knowledge, omitted rules, and Q1 reconstruction error change that trade-off?

End-to-end Q2 checks answer a narrower engineering question: can an approved view still be delivered through the existing proof of concept, and at what measured overhead?

A. Exact Micro-Worlds

The first tier enumerates all worlds and bounded query sequences for small graphs. Scenario families cover direct requests, multi-hop ontology inference, the two-release mosaic in Figure 1, overlapping-query differencing, refusal leakage, correlated attributes, and two-identity transcript pooling. Exhaustive comparison with an exact safe oracle tests whether the implemented gate matches the declared semantics. “No breach” in this tier means no breach within the enumerated graph, rule, observable, and horizon bounds—not universal privacy.

TABLE I
PRIMARY DISCLOSURE MECHANISMS.

Mechanism	Decision rule
Q2 selector	Fixed predicate/profile selection followed by encrypted delivery; no history or semantic gate.
Explicit redaction	Removes named protected triples but does not test derived knowledge.
Task-minimal	Minimizes the current response without inference or history checks.
Stateless censor	Tests the current response semantically but discards prior exchanges.
Stateful logical	Preserves the declared logical-opacity invariant over history.
Hybrid gate	Combines logical opacity, probabilistic charging, and a coalition transcript.
Exact safe oracle	Exhaustive highest-utility safe response in bounded micro-worlds.

B. Generated Laboratory Episodes

The main empirical tier uses the companion Generation environment to instantiate independent laboratories, members, projects, papers, datasets, instruments, collaborators, funding, and unpublished outcomes. The authoritative simulator state supplies protected alternatives and oracle graphs. Mailboxes supply the evidence available to personal agents, producing reconstructed graphs for the Q1 sensitivity condition.

Legitimate requests include finding expertise, locating a shareable instrument or dataset, identifying a potential collaborator, checking whether public prior work meets a requirement, and obtaining a generalized project status. Protected families include unpublished results, membership, pending patents, restricted partnerships, locations or times whose conjunction is sensitive, and relationships derivable only through graph paths.

C. Mechanisms and Attackers

Table I keeps the main comparison compact. The exact safe oracle is a reference bound in enumerable worlds, not a deployable baseline.

Attackers include a fixed requester, a greedy information-gain policy, a symbolic multi-hop planner, a Bayesian expected-information-gain policy, an adaptive differencing requester, and two colluding identities. An LLM red team is included as an ecological stress test. Failure of a tested attacker is empirical evidence only; it is never substituted for the declared guarantee.

D. Metrics and Analysis

The primary logical metric is the fraction of initially hidden protected propositions newly entailed by the final transcript:

$$\text{LBR} = \frac{|(\text{Cn}_\Sigma(K_B \cup H_T) \setminus \text{Cn}_\Sigma(K_B)) \cap \mathcal{P}|}{|\mathcal{P} \setminus \text{Cn}_\Sigma(K_B)|}. \quad (9)$$

We also report worst empirical cumulative log-odds amplification, attacker success at a preregistered posterior threshold, rounds to first breach, task completion and answer correctness, excess disclosure, refusal and generalization rates, safe-oracle regret, gate latency, and end-to-end Q2 latency.

An interaction episode is the replication unit. Mechanisms are paired on identical worlds, tasks, protected alternatives, candidate pools, requester seeds, and horizons. We split scenario templates—not individual questions—between development

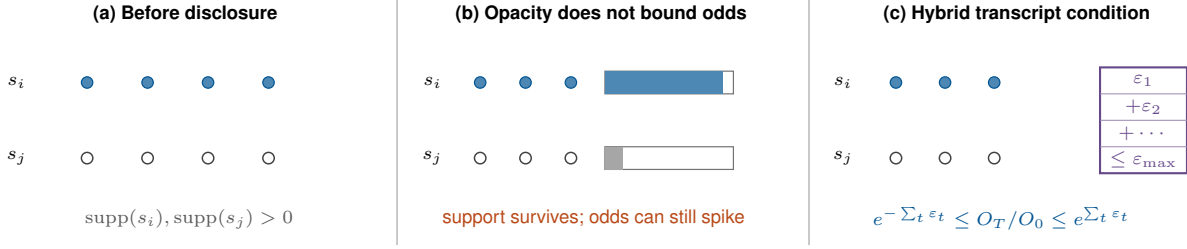


Fig. 4. Logical and probabilistic confidentiality address different failures. Possible-world opacity preserves support for each protected alternative, while the likelihood-ratio condition bounds the change in posterior odds O_T relative to prior odds O_0 . Dots indicate support only; the mass bars are schematic, not world counts or experimental data. The hybrid condition does not force a balanced posterior.

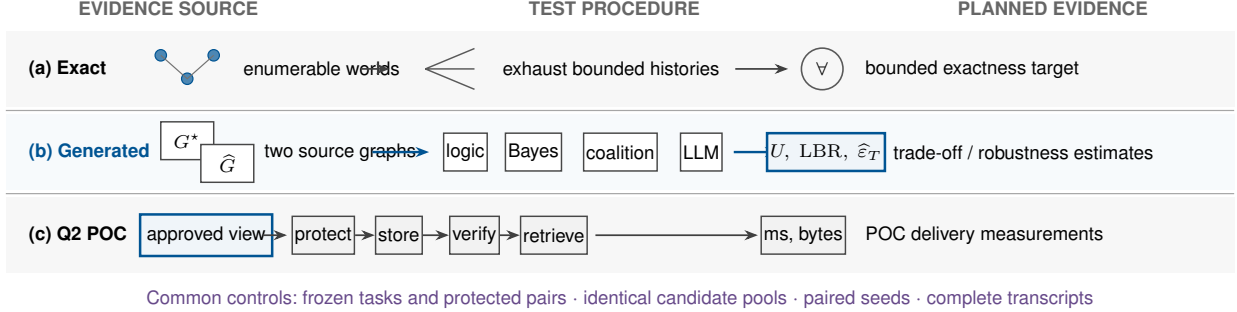


Fig. 5. The planned evaluation separates three kinds of evidence. Exhaustive micro-worlds target the declared semantics, generated laboratory worlds estimate adaptive privacy–utility behavior and Q1 sensitivity, and Q2 integration measures only proof-of-concept delivery feasibility.

and held-out evaluation. Exact counts are separated from stochastic attack estimates, which receive paired uncertainty intervals. Privacy–utility curves provide the primary comparison.

VI. RESULTS PLAN

This section is intentionally a reporting shell. It will be completed only after the threat model, rule language, candidate generator, baselines, splits, and primary metrics are frozen. No empirical figure is rendered before measured data exist; the planned privacy–utility and longitudinal plots are specified in `FIGURE_PLAN.md`.

A. Authorized Inference and the Privacy–Utility Frontier

[Report the prevalence, attack class, and median round of first protected inference under the existing Q2 selector. Then compare the stateful gate with Q2, task-minimal disclosure, and the stateless censor on held-out episodes. Lead with a privacy metric and its paired utility cost.]

B. Longitudinal, Coalition, and Model Robustness

[Report leakage or attack success against interaction horizon and coalition size. Then report a two-dimensional robustness analysis over hidden background knowledge and omitted inference rules. Separate guarantee violations caused by checker defects from failures outside the declared model.]

The oracle-versus-reconstructed comparison will be reported separately because it answers a Q1 sensitivity question rather

than a Q3 mechanism comparison. Its key quantities are source-graph F1, task utility, logical breach, and attack success on paired episodes.

C. Q2 Integration Evidence

These measurements establish only whether the approved semantic response can traverse the existing Q2 path at proof-of-concept scale. They are not evidence of production identity assurance, availability, revocation, or Byzantine fault tolerance.

VII. RELATED WORK

A. Selective Disclosure and Attribute-Gated Encryption

Ciphertext-policy attribute-based encryption determines which credential combinations can recover a released plaintext. Multi-authority designs distribute attribute issuance across administrative domains [1]. This is complementary to the present problem: encryption protects a fragment before decryption, whereas semantic control constrains what an authorized recipient can learn after decryption.

B. Controlled Query Evaluation

Controlled Query Evaluation filters or alters database answers to protect declared secrets [7], [10]. Instance-indistinguishability semantics preserve a secret-free alternative [11]; dynamic CQE makes the answer depend on the previous query sequence [8]; epistemic policies constrain what a requester may come to know [9]. These works provide the formal backbone for our logical layer. Our experimental setting adds subgraph-valued and agent-proposed responses, an explicit

TABLE II
MAIN HELD-OUT PRIVACY-UTILITY RESULTS. EVERY ENTRY REMAINS A PLACEHOLDER UNTIL THE EXPERIMENT IS FROZEN AND EXECUTED.

Mechanism	Task utility $\uparrow$	LBR $\downarrow$	Empirical $\widehat{\epsilon}_T$ $\downarrow$	Attack success $\downarrow$	Excess disclosure $\downarrow$	Gate latency $\downarrow$
Q2 selector	[value]	[value]	[value]	[value]	[value]	[value]
Task-minimal	[value]	[value]	[value]	[value]	[value]	[value]
Stateless censor	[value]	[value]	[value]	[value]	[value]	[value]
Stateful logical	[value]	[value]	[value]	[value]	[value]	[value]
Hybrid gate	[value]	[value]	[value]	[value]	[value]	[value]
Exact safe oracle	[value]	[value]	[value]	[value]	[value]	[value]

TABLE III
END-TO-END PROOF-OF-CONCEPT CHECKS.

Check or measurement	Result
Authorized recipient decrypts	[pass rate]
Missing required attribute is denied	[pass rate]
Ciphertext/blob tampering is detected	[pass rate]
Cross-lab publish/retrieve succeeds	[pass rate]
Semantic-gate latency	[value]
End-to-end latency and expansion	[value]

correlated-inference layer, coalition histories, and an encrypted cross-lab delivery substrate.

C. Correlated and Graph Inference

Pufferfish privacy makes secret pairs and a family of data-generating distributions explicit [5]. This is attractive for correlated graph facts, but one-shot mechanisms do not generally compose without stronger conditions [6]. Differential privacy provides the canonical likelihood-ratio and composition perspective [12], although our unit of protection is a declared semantic alternative rather than a neighboring row. Probabilistic inference-proof filtering shows why logical non-entailment can coexist with high posterior confidence [13]. Graph topology and public attributes can also reveal concealed relationships [14]. We treat exact deduction, correlated inference, and learned attack success as distinct evidence.

VIII. DISCUSSION AND LIMITATIONS

The guarantee is declared, not universal. Logical opacity is only as complete as Σ , K_B , the protected policy, and the observable model. The probabilistic condition is only as credible as Θ and its likelihood bounds. Robustness experiments deliberately hide background facts and rules from A to locate this boundary rather than conceal it.

Refusal is an output. A system that releases only "safe" plaintext can still leak through which questions are refused, how specifically they are answered, or what metadata accompanies a release. The modeled transcript must therefore include the mechanism's response mode. Timing and external network behavior are included only when explicitly modeled.

A language model is not the proof. An LLM may interpret a task, propose candidates, or search for attacks. It cannot certify its own answer. The security claim comes from the explicit rule checker, protected-alternative model, probabilistic bound, and persistent ledger. An LLM red team expands empirical coverage but cannot prove the absence of an attack.

Q1 error and Q3 failure must remain distinguishable. Oracle graphs isolate the disclosure mechanism. Reconstructed graphs test end-to-end sensitivity, including missing, spurious, conflicting, or poorly resolved facts. A leak caused by an inaccurate source graph has a different remediation from a gate that violates its declared semantics.

The system is a research proof of concept. The work seeks formal and empirical results, not a production platform. Current identity registration, credential handoff, revocation, storage availability, endpoint security, and validator configuration require hardening before deployment. Those limitations bound the systems claim but do not erase the semantic problem: perfect transport security would still not prevent the attack in Figure 1.

IX. CONCLUSION

Access control answers whether Agent B may decrypt a released fragment. It does not answer what B can infer after combining that fragment with prior exchanges and background knowledge. This paper makes the accumulated transcript the security object and places a stateful semantic gate between graph construction and encrypted delivery. The gate preserves declared logical alternatives, meters posterior-odds amplification, and chooses the most useful feasible response from a transformation ladder. The planned evaluation separates bounded exact guarantees, empirical adaptive attacks, Q1 reconstruction sensitivity, and Q2 delivery feasibility. *[Replace this final sentence with the measured privacy-utility conclusion after results are available.]*

APPENDIX A PROOF AND MODEL CHECKLIST

The final supplementary material will contain: (i) the syntax and semantics of the chosen Horn, Datalog, or description-logic fragment; (ii) the formal mapping from graph, text, refusal, and metadata observations into H_t ; (iii) the inductive logical-opacity proof; (iv) the conditional-history likelihood-ratio composition proof; (v) the treatment of nonzero δ ; and (vi) an exhaustive equivalence check between the implementation and possible-world semantics on micro-worlds.

APPENDIX B ATTACK AND SCENARIO INVENTORY

The frozen artifact will enumerate direct protected requests, multi-hop path completion, individually safe releases that compose, overlapping-query differencing, correlated link and

attribute prediction, refusal and response-mode leakage, version and metadata inference, coalition pooling, and Sybil-style history splitting under the declared identity model. Each scenario will specify the legitimate task, protected alternatives, B’s initial knowledge, permitted rules, candidate answers, complete observables, and expected exact-safe action.

APPENDIX C

REPRODUCIBILITY ARTIFACTS

The release will include graph worlds, mailbox traces, oracle and reconstructed exports, ontologies and rules, protected-pair policies, task answer keys, candidate pools, attacker policies, coalition mappings, random seeds, complete transcripts, privacy ledgers, checker justifications, environment manifests, and analysis scripts. Every reported privacy result will identify the exact observable model and attacker class under which it was obtained.

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